

On the Stiffness Selection for Torque Controlled Series-Elastic Actuators

Wesley Roozing¹, Jörn Malzahn¹, Navvab Kashiri¹, Darwin G. Caldwell¹ and Nikos G. Tsagarakis¹

Abstract—Series-elastic actuators are quickly becoming the core component of robots operating in real world environments, allowing for robust, safe, torque controlled robots. This paper investigates the influence of the selected stiffness and control parameters. By consolidating several analyses, it is shown two opposing criteria exist for the choice of stiffness: lowering stiffness increases actuator transparency, while increasing stiffness increases torque tracking bandwidth. This fact is used in the context of impedance rendering to propose an optimal stiffness selection criterion that maximises the renderable impedance frequency. This provides a concrete method for choosing stiffness based on the application of the joint. Extensive simulation and experimental results are presented that validate the analyses.

Index Terms—Force Control, Compliance and Impedance Control, Compliant Joint/Mechanism.

I. INTRODUCTION

IN recent years robotics has shifted its actuation approach from position controlled and high intrinsic stiffness drives, to intentionally compliant, force controlled actuation units. This paradigm shift has sparked significant discussion on how to design and control these actuators. Particularly, the point at which a stiff actuator becomes a compliant actuator is not clear, and what level of compliance is suitable and beneficial for a given application is still an open research question. In a sense, any actuator can be considered compliant due to the finite rigidity of its structural components; indeed, by measuring the deformations of its internal components, even a “stiff” actuator can be seen as a compliant actuator that can be force controlled [1]–[3]. In recent literature the characterization of an actuator as compliant and its distinction from a stiff drive is mostly based on the intentional addition of compliance within the transmission system of the actuator. Actuators with added constant series compliance have been commonly known as Series Elastic Actuators (SEAs). The concept of SEAs was first introduced in the work of Pratt et al. [4], and have seen significant adoption in recent years. Throughout these works, very different stiffness values have been adopted, ranging roughly from 100 to 50,000 Nm/rad.

Manuscript received: February, 15th, 2017; Revised May, 22nd, 2017; Revised June, 27th, 2017; Accepted July, 4th, 2017.

This paper was recommended for publication by Editor Han Ding upon evaluation of the Associate Editor and Reviewers’ comments. This work is supported by the WALK-MAN FP7-ICT 611832 European Commission project.

¹The authors are with the Department of Advanced Robotics, (Fondazione) Istituto Italiano di Tecnologia, via Morego, 30, 16163 Genova, Italy {wesley.roozing, jorn.malzahn, navvab.kashiri, darwin.caldwell, nikos.tsagarakis}@iit.it

Digital Object Identifier (DOI): see top of this page.

Some works that consider stiffness selection for SEAs exist. In [5] a dimensionless analysis was presented that investigates the influence of system parameters on torque control performance and output impedance. [6], [7] demonstrated how the presence of compliance enables stable force control in uncertain environments. The influence of load motion on the torque tracking performance was investigated in [8]. In terms of impedance rendering, [3], [9] suggested that if passive operation is desired the selected mechanical stiffness should be higher than the rendered stiffness.

Important applications of force controlled actuators include interaction with humans and/or the environment, such as in rehabilitation [9] or assistive [11] devices, humanoids and field robots [12]–[14], [16] meant to operate in partially known and unstructured workspaces, where unexpected collisions are common. Therefore, output impedance over the entire frequency range, not only the controllable torque bandwidth, is an important aspect of these actuators.

Conventional performance metrics for stiff actuators, such as closed-loop tracking bandwidth, do not seem as useful for compliant actuators. For example, the torque tracking bandwidth, commonly defined in fixed-output configuration, conveys only limited information about the output impedance or interaction performance of an actuator. Interaction performance can be defined as the actuator’s ability to track desired torques (which may be zero for transparent operation) in the presence of output motion.

This paper investigates the influence of the selected stiffness level and control parameters in a compliant actuation unit. The main contribution is a novel method for optimal stiffness selection, relying upon the trade-off between torque tracking bandwidth and transparency, thereby maximising the frequency up to which an impedance can be rendered accurately. To this end, it consolidates several analyses on SEA dynamics and control including the effects of active/passive compliance on the performance of actuation drives in terms of tracking zero and non-zero torques in the presence of load motion, as well as active impedance rendering. To validate the analyses, we present simulations and experimental results.

The paper is organised as follows. Sec. II and III introduce the model and its analysis. A stiffness selection guideline for impedance rendering is proposed in Sec. IV, followed by simulations and experimental results in Sec. V and VI. Concluding remarks are reported in Sec. VII.

II. MODELLING

Fig. 1 shows the linear dynamics model of a general SEA. The model consists of motor rotor and gearbox inertiae I_m ,

I_g , coupled through a spring-damper element with stiffness k_g and internal damping d_{mg} , modelling the finite rigidity of the Harmonic Drive gearbox. The gearbox inertia and load inertia I_l are coupled through the flexible transmission element, modelled as a spring-damper element with stiffness and internal damping k_b and d_{gl} . The inertiae experience viscous friction with the environment through bearings, with respective parameters d_m, d_g, d_l . We denote their positions as q_m, q_g, q_l respectively, and define the torques acting on the motor rotor due to winding current and on the load through environmental interaction as τ_m and τ_e respectively.

The equations of motion for the joint model of Fig. 1 are derived from the torque balance of each individual inertia:

$$\begin{aligned} I_m \ddot{q}_m &= \tau_m - d_m \dot{q}_m + d_{mg}(\dot{q}_g - \dot{q}_m) + k_g(q_g - q_m) \\ I_g \ddot{q}_g &= -d_g \dot{q}_g - d_{mg}(\dot{q}_g - \dot{q}_m) - k_g(q_g - q_m) \\ &\quad + d_{gl}(\dot{q}_l - \dot{q}_g) + k_b(q_l - q_g) \\ I_l \ddot{q}_l &= \tau_e - d_l \dot{q}_l - d_{gl}(\dot{q}_l - \dot{q}_g) - k_b(q_l - q_g), \end{aligned} \quad (1)$$

where τ_m is the torque generated by the winding current, which together with the motor-side inertia and damping parameters is reflected through the gearbox ratio. We explicitly define the deflection variable $\Delta = q_g - q_l$, and the torque transmitted to the load $\tau = k_b \Delta + d_{gl} \dot{\Delta}$. In estimating the torque for control, the damping component d_{gl} is usually ignored, i.e. $\tau \approx k_b \Delta$, due to the very low damping properties of metal elastic elements usually employed.

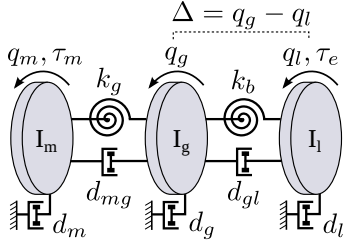


Fig. 1: Linear Dynamics Model.

III. LINEAR ANALYSIS

A. Simplification

For most SEAs, the Harmonic Drive can be considered significantly stiffer than the flexible transmission element. Therefore, a significant simplification can be obtained by considering the Harmonic Drive to be rigid. By setting $q_m \equiv q_g$, $\dot{q}_m \equiv \dot{q}_g$, and $\ddot{q}_m \equiv \ddot{q}_g$ in (1), the equations simplify to:

$$(I_m + I_g) \ddot{q}_m = \tau_m + k_b(q_l - q_m) + d_{gl}(\dot{q}_l - \dot{q}_m) - (d_m + d_g) \dot{q}_m, \quad (2)$$

$$I_l \ddot{q}_l = \tau_e - k_b(q_l - q_m) - d_{gl}(\dot{q}_l - \dot{q}_m) - d_l \dot{q}_l. \quad (3)$$

The simplified model (2)–(3) is 4th order, and therefore significantly simpler to analyse. However, even at frequencies lower than the resonance frequency of the gearbox, the associated phase lag can result in closed-loop instability. Therefore, we always use the full model (1) for simulation.

B. Torque Derivation

Disregarding the load-side dynamics for a moment, we take the Laplace transform of (2) and rewrite in terms of deflection $q_g(s) - q_l(s) := \Delta(s)$:

$$[Is^2 + (d_M + d_{gl})s + k_b] \Delta(s) = -[Is^2 + d_M s] q_l(s) + \tau_m(s), \quad (4)$$

where now $I = I_m + I_g$ and $d_M = d_m + d_g$. Assuming a feedback controller in the form $C(s) = K_p + K_d s$ (which measures deflection and uses $\tau(s) \approx k_b \Delta(s)$) with feed-forward term $\lambda(s)$:

$$\tau_m(s) = C(s) [\tau^*(s) - k_b \Delta(s)] + \lambda(s) \tau^*(s), \quad (5)$$

and substituting into (4), we obtain

$$[Is^2 + (d_M + d_{gl})s + k_b(1 + C(s))] \Delta(s) = -[Is^2 + d_M s] q_l(s) + [C(s) + \lambda(s)] \tau^*(s), \quad (6)$$

where $\tau^*(s)$ denotes the desired torque in the Laplace domain. Considering that $\tau(s) = (k_b + d_{gl}s) \Delta(s)$, (6) can be solved for $\tau(s)$ as the sum of input responses from $\tau^*(s)$ and $q_l(s)$:

$$\tau(s) = (k_b + d_{gl}s) [\Delta_{\tau^*}(s) \tau^*(s) + \Delta_{q_l}(s) q_l(s)], \quad (7)$$

where

$$\begin{aligned} \Delta_{\tau^*}(s) &= \frac{\Delta(s)}{\tau^*(s)} = \frac{K_d s + K_p + \lambda(s)}{Is^2 + D_\Delta s + K_\Delta}, \\ \Delta_{q_l}(s) &= \frac{\Delta(s)}{q_l(s)} = \frac{-(Is^2 + d_M s)}{Is^2 + D_\Delta s + K_\Delta}, \end{aligned} \quad (8)$$

and

$$\begin{aligned} D_\Delta &= k_b K_d + d_M + d_{gl}, \\ K_\Delta &= k_b (K_p + 1). \end{aligned} \quad (9)$$

Note that if $q_l(s)$ is chosen as the load-side dynamics (3), the derivation yields $\tau(s)$ with inputs $\tau^*(s)$ and $\tau_e(s)$. However, instead of choosing a predefined impedance, Eq. (7) is more general, and useful for obtaining the robust torque control performance in the presence of arbitrary output motion, as well as assessing actuator transparency. In Sec. III-D and III-E we shall investigate this dichotomy, in the form of the two boundary cases of (7): when $\tau^*(s) \equiv 0$ (the actuator should be transparent) and when $q_l(s) \equiv 0$ (the load is fixed or sufficiently large). A similar analysis was performed in [5], however we report some additional insights with regards to the transfer functions zeroes. First, we briefly discuss the procedure for pole placement that governs both.

C. Control: Pole Placement Procedure

Inspecting (6)–(8), we observe that $\Delta_{\tau^*}(s)$ and $\Delta_{q_l}(s)$ have equal denominators. Consequently, an appropriate choice of $C(s)$ allows us to arbitrarily place the closed-loop poles for both, by selection of K_p, K_d . Using the characteristic polynomial of a general 2nd order system:

$$s^2 + 2\zeta \omega_0 s + \omega_0^2 = 0, \quad (10)$$

where ω_0 is the characteristic frequency and ζ is the damping ratio, allows us to place the poles of the closed-loop system

in desired locations. Writing the denominators of (8) in the form of (10) and solving for K_d , we obtain the controller derivative gain as a function of the closed-loop damping ratio ζ and proportional gain K_p :

$$K_d = \frac{2 I \omega_0 \zeta - (d_M + d_{gl})}{k_b}, \quad (11)$$

where

$$\omega_0 = \sqrt{K_\Delta / I} = \sqrt{k_b (K_p + 1) / I}. \quad (12)$$

Using (11) and choosing $\zeta = 1$, the gain selection is simplified to selection of only the proportional gain K_p , which is now equivalent to selection of the pole frequency ω_0 (solving (12) for K_p):

$$K_p = \frac{\omega_0^2 I}{k_b} - 1 = \frac{\omega_0^2}{\omega_n^2} - 1, \quad (13)$$

where the locked-output¹ plant natural resonance frequency $\omega_n = \sqrt{k_b / I}$.

Choosing the closed-loop poles, that define the control bandwidth, as a constant times the resonance frequency:

$$\omega_0 = \alpha \omega_n, \quad (14)$$

results in a simple tuning rule with α as the only parameter, which is independent of stiffness: $K_p = \alpha^2 - 1$. Note that $K_p > 0$ requires $\omega_0 > \omega_n$, or equivalently, $\alpha > 1.0$. Similarly, we constrain $K_d \geq 0$. Lastly, from the two-input system described by (7)–(9), one can observe that the feed-forward gain $\lambda(s)$ affects $\Delta_{\tau^*}(s)$, while $\Delta_{q_l}(s)$ is independent of $\lambda(s)$. Hence, we proceed the analysis with $\lambda(s) \equiv 1$, resulting in zero steady-state error for $\Delta_{\tau^*}(s)$.

D. Transparency: Load Motion to Residual Torque

The transparency of an actuator in this context is defined as the ability to track zero torque in the presence of output motion; transparency is the inverse of the residual torques given an imposed output motion.

Setting $\tau^*(s) \equiv 0$ in (7) and solving for $\frac{\tau(s)}{q_l(s)}$, we obtain

$$\frac{\tau(s)}{q_l(s)} = \frac{-(k_b + d_{gl} s)(I s^2 + d_M s)}{I s^2 + D_\Delta s + K_\Delta}. \quad (15)$$

We hereafter refer to (15) as the *transparency function* of the actuator. Using $s = j\omega$ with $j = \sqrt{-1}$, we find the high frequency gain of (15) is given by $-(k_b + d_{gl} s)$, i.e. the passive impedance of the flexible element. Hence, the high frequency transparency is upper bounded by the flexible element stiffness, since it is reasonable to assume d_{gl} to be a function of k_b due to the mechanical implementation. However, since for $\omega \rightarrow 0$ we have (15) $\rightarrow 0$ (applying the final value theorem), we find that in principle the actuator can be made arbitrarily transparent at a given frequency by placing the poles at an arbitrarily high frequency.

The transparency function shows that the transparency of the actuator is determined by 1) the frequency ω_0 of the closed-loop poles and 2) the stiffness k_b . Fig. 2 shows the effect of varying both these parameters.

¹For a non-locked output the natural frequency is equal to $\sqrt{\frac{k_b(I+I_l)}{I I_l}}$; while for a locked output, i.e. $I_l \gg I$, this converges to $\sqrt{k_b / I}$.

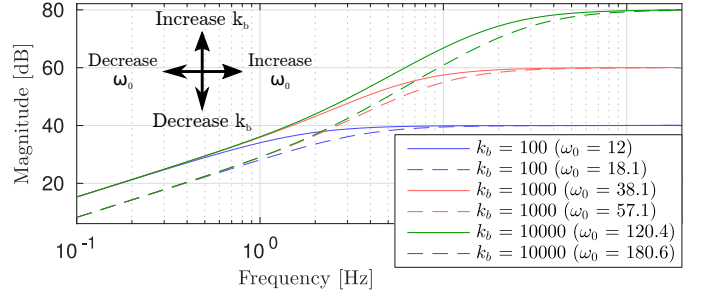


Fig. 2: Varying k_b and ω_0 in (15): Residual torque from load motion for various stiffness values and closed-loop poles. Here we have chosen $\alpha = 1.2$ and 1.8 for each stiffness value.

At low frequency the transparency depends only on α (i.e. control gain), whereas at high frequency it depends only on k_b , with a transition period in between. At high frequencies each actuator converges to a gain of k_b , with lower stiffness actuators showing lower residual torques, i.e. higher transparency. Furthermore, Fig. 2 shows that increased control bandwidth improves the transparency; however, such an increase for stiffer actuators only results in surpassing the transparency of lower stiffness actuators up to a limited frequency, as the transparency of stiffer actuators intrinsically deteriorates at higher frequencies due to the upper bound defined by increased stiffness. High frequency transparency is however important in impacts due to their impulsive nature; high stiffness SEAs may provide insufficient protection against impacts, despite high control bandwidth.

Consider two actuators with different stiffness values $k_{b,1}$ and $k_{b,2}$, with the closed-loop pole frequency of the first actuator given by ω_1 . By calculating the intersection between the extrapolation of the +40 dB/dec slope of the first actuator and the stiffness of the second actuator in dB, the pole frequency ω_2 required for the second actuator to obtain the same transparency (up to ω_1) can be derived:

$$\omega_2 = \exp \left[\frac{1}{2} \ln \left(\frac{k_{b,2}}{k_{b,1}} \right) + \ln(\omega_1) \right] = \sqrt{\frac{k_{b,2}}{k_{b,1}}} \omega_1, \quad (16)$$

which, by substituting $\omega_1 = \alpha_1 \sqrt{\frac{k_{b,1}}{I}}$ and $\omega_2 = \alpha_2 \sqrt{\frac{k_{b,2}}{I}}$, gives $\alpha_1 = \alpha_2$, confirming the equal low frequency transparency for constant values of α in Fig. 2. In other words, choosing α enables us to choose the low frequency transparency independently of stiffness.

Inspecting the numerator of (15), the first zero (other than $s = 0$) is located at $s = -d_M / I$, which corresponds to the intuition that lowering the motor-side damping increases the transparency. Apart from clever mechanical design to minimise friction, active solutions have been proposed, such as friction compensators, and more recently, disturbance observers [11], [15]. The presence of s in each term reveals the velocity dependence, and in fact feeding back the link velocity signal can effectively remove the zero at $s = -d_M / I$ in velocity sourced actuators [8]. Hereafter the zero at $s = -k_b / d_{gl}$ will be disregarded, since due to $k_b \gg d_{gl}$ it falls outside the frequency range of interest.

E. Torque Tracking

Setting $q_l(s) \equiv 0$ in (7) and solving for $\frac{\tau(s)}{\tau^*(s)}$, we obtain

$$\frac{\tau(s)}{\tau^*(s)} = \frac{(k_b + d_{gl} s)(K_d s + K_p + 1)}{I s^2 + D_\Delta s + K_\Delta}. \quad (17)$$

We hereafter refer to (17) as the *torque tracking function* of the actuator. Eq. (17) has a denominator identical to that of (15) which allows for pole placement as presented in Sec. III-C, however in (17) the control gains also affect the closed-loop zeroes. Fig. 3 shows the effect of varying both k_b and ω_0 in (17). The low frequency zero in (17) is given by

$$s = -\frac{K_p + 1}{K_d} = -\frac{I \omega_0^2}{2 I \omega_0 \zeta - (d_M + d_{gl})}. \quad (18)$$

For $\omega_0 > 0$, this zero quickly converges to $\approx \omega_0/(2\zeta)$; as the chosen pole frequency increases, the distance between this zero and the poles increases, which may result in a ripple in the torque response despite critical damping of the poles.

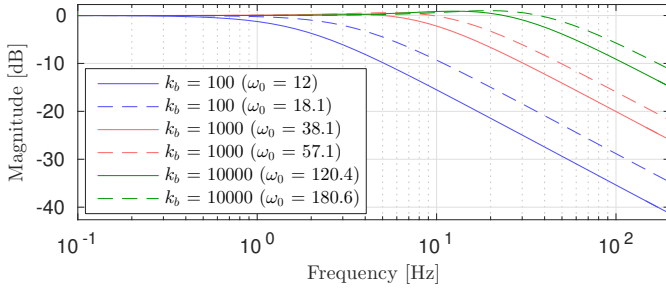


Fig. 3: Varying k_b and ω_0 in (17): Torque tracking. Here we have chosen $\alpha = 1.2$ and 1.8 for each stiffness value.

F. Motor Power

An oft-cited disadvantage of SEAs with lower stiffness is higher power consumption due to the larger motor movements required to achieve the deflection needed for a certain desired torque. To investigate this, we use (4) in the same boundary case as Sec. III-E with $\tau(s) = (k_b + d_{gl} s) \Delta(s)$:

$$\left. \frac{\tau_m(s)}{\tau(s)} \right|_{q_l(s) \equiv 0} = \frac{I s^2 + (d_M + d_{gl}) s + k_b}{k_b + d_{gl} s}, \quad (19)$$

which can be observed (Fig. 4) to increase with +40 dB/dec after the mechanical resonance now present in the numerator. Due to input torque limitation arising from maximum allowable current, further exasperated by a reduction in torque resulting from the back-EMF component and operating voltage, achievable motor power is limited. Thus, the mechanical resonance frequency is an inherent source of torque tracking bandwidth limitation, regardless of control strategy [5]. In the other boundary case (transparency, Sec. III-D), the deflection is ideally zero and the required motor torque difference between actuators of different stiffness values disappears, i.e. becomes independent of k_b . This is shown by solving (4) for $\frac{\tau_m(s)}{q_l(s)}$ while setting $\Delta(s) \equiv 0$:

$$\left. \frac{\tau_m(s)}{q_l(s)} \right|_{\Delta(s) \equiv 0} = I s^2 + d_M s, \quad (20)$$

which depends only on I and d_M . In conclusion, the motor torque for a lower stiffness SEA is much higher while tracking

torque profiles above the mechanical resonance. However, this is not the case for lower frequencies or interaction with small torques, which require approximately identical power regardless of stiffness.

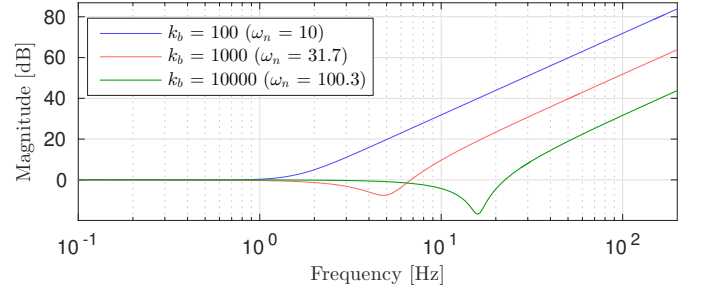


Fig. 4: Motor torque as function of delivered output torque.

IV. OPTIMAL STIFFNESS SELECTION FOR IMPEDANCE RENDERING

A. Impedance Rendering and Motion Control

In many real-world applications the torque control loop is supplemented by an outer loop to provide motion control and impedance rendering to the robotic system. The desired torque is commonly given by a 1st order impedance:

$$\tau^* = K (q_l^0 - q_l) + D (\dot{q}_l^0 - \dot{q}_l), \quad (21)$$

where $q_l^0, \dot{q}_l^0$ denote the reference position of the link and its derivative, and K, D denote stiffness in Nm/rad and damping in Nms/rad, respectively. Given a stable torque control loop and $K_d \geq 0$, both the zeroes and the poles of the inner loop will be in the left-hand complex plane, and as such for $K, D \geq 0$ the entire closed-loop system will be asymptotically stable. However, it is well known that in practical systems with nonlinear effects like friction, discrete-time control, delays, filtering, noise and quantisation, increasing the gains of the inner (torque) loop can have a detrimental effect on the stability of the overall system, despite the inner loops themselves being stable. This results in a limitation on the achievable closed-loop pole frequency (Sec. III-C), and therefore stiffness selection requires careful consideration. For an elaborate discussion and experimental results, see e.g. Focchi et al. [16]. To investigate impedance rendering, we set $q_l^0 \equiv \dot{q}_l^0 \equiv 0$ in (21) and transform into the Laplace domain:

$$\tau^*(s) = -(K + D s) q_l(s). \quad (22)$$

Substituting (22) into (7) and solving for the rendered impedance $Z(s)$, we obtain:

$$\begin{aligned} Z(s) &= \frac{\tau(s)}{q_l(s)} \\ &= (k_b + d_{gl} s) [-(K + D s) \Delta_{\tau^*}(s) + \Delta_{q_l}(s)] q_l(s) \\ &= -(k_b + d_{gl} s) \frac{a s^2 + b s + c}{I s^2 + d s + e}, \end{aligned} \quad (23)$$

with

$$\begin{aligned} a &= I + K_d D, & d &= k_b K_d + d_M + d_{gl}, \\ b &= D (K_p + 1) + K_d K + d_M, & e &= k_b (K_p + 1), \\ c &= K (K_p + 1), \end{aligned} \quad (24)$$

which is functionally equal to Fig. 5. Contrary to pure torque tracking where the objective is tracking torque independent of the load motion, the objective here is accurate rendering of desired output impedance $Z(s) = \tau(s)/q_l(s)$; perfect rendering would imply $Z(s) = -K - Ds$.

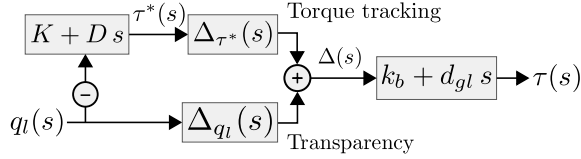


Fig. 5: Impedance rendering block diagram.

The structure of (23) suggests both the torque tracking bandwidth and transparency function can limit the impedance rendering capability of an actuator. We demonstrate this with an example of two actuators with stiffness values of 10000 and 100 Nm/rad respectively, both rendering a reference impedance of $K = 1000$ Nm/rad and $D = 20$ Nms/rad. The former case is shown in Fig. 6, which displays a plot of the rendered impedance (23), a reference impedance $K + Ds$, the transparency function (15), and the torque tracking function (17). Fig. 6 reveals that the actuator transparency is a limiting factor; the minimum renderable impedance is dictated by the transparency function.

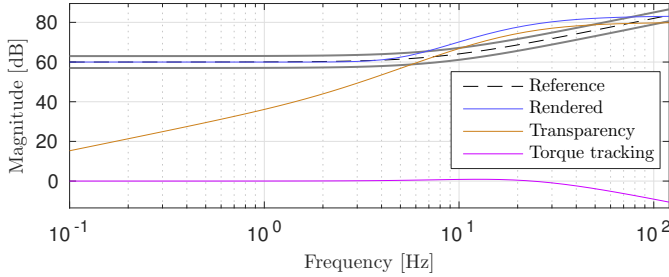


Fig. 6: Impedance rendering for $k_b = 10000$ Nm/rad, limited by the transparency function. Here $K = 1000$ and $D = 20$.

Conversely, Fig. 7 shows the same reference impedance rendered by the lower stiffness actuator. In this case, the substantially reduced torque tracking bandwidth is the limiting factor to the impedance rendering bandwidth. In both figures, the solid grey lines denote the ± 3 dB margin.

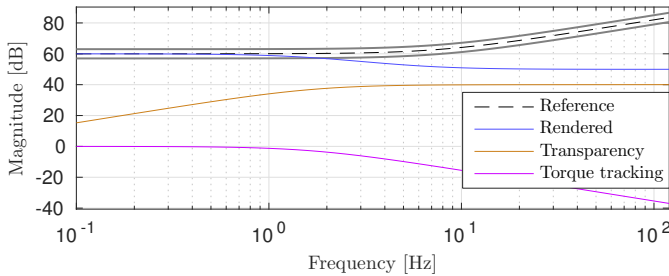


Fig. 7: Impedance rendering for $k_b = 100$ Nm/rad, limited by the torque tracking bandwidth. Here $K = 1000$ and $D = 20$.

Setting $K, D = 0$ in (23) immediately reveals similarity to (15). It can also be seen that for small K, D , (23) tends to (15) at high frequencies, with an increase of gain resulting from $K_d D$ in a of (23).

B. Optimal Stiffness Selection

Following the presented analyses, we can suggest an optimal stiffness selection guideline for SEAs. Choice of rotor, its inertia, torque constant and gear ratio are largely determined by the application torque and speed requirements, and since their choice is the same as for stiff motors, we will not discuss them here. Sec. III showed that two opposing criteria exist for the choice of stiffness: Lowering k_b increases transparency, and increasing k_b increases torque tracking bandwidth. Using the context of impedance rendering, we can frame these two criteria into a single metric that can be optimised on, as will be demonstrated in the following.

For a robotic system under motion control, three operating parameters can be defined:

- 1) the minimum/maximum load inertia I_l^{min}, I_l^{max} that will be seen by the actuator, and
- 2) the motion bandwidth requirement ω_m of the robot.

These parameters define the range of impedance values that have to be replicated by the actuator. Given a frictionless joint with load I_l , the joint impedance that places the poles at a frequency ω_m is given by $Z_{ref}(s) = K + Ds$, with

$$K = \omega_m^2 I_l, \quad D = 2 I_l \omega_m \zeta, \quad (25)$$

where we set $\zeta = 1$. Both K and D are linear in I_l , so that the required minimally renderable impedance $Z_{min}(s)$ of the actuator is given by (25) with $I_l = I_l^{min}$.

By defining a margin of acceptable accuracy ϵ , we can optimise for the maximum frequency ω_b up to which a reference impedance $Z_{ref}(s)$ can be rendered accurately:

$$\max_{k_b} \omega_b \quad \text{s.t.} \quad \frac{|Z(j\omega)|}{|Z_{ref}(j\omega)|} \in [-\epsilon, +\epsilon] \quad \forall \omega \leq \omega_b, \quad (26)$$

where $j = \sqrt{-1}$. Given I_l^{min}, I_l^{max} and ω_m , for the motion requirements to be fulfilled we require

$$\omega_b \geq \omega_m \quad \forall I_l \in [I_l^{min}, I_l^{max}], \quad (27)$$

i.e. the impedance can be rendered accurately up to at least the motion bandwidth, for the entire link inertia range.

The optimisation (26) and impedance replication requirement (27) provide a concrete method for choosing k_b based on the application of the joint. Fig. 8 shows an example of the application of (26) for a reference impedance of $K = 1000$ Nm/rad and $D = 20$ Nms/rad, resulting in an optimal stiffness of $k_b = 752$ Nm/rad and a rendering bandwidth of ≈ 12 Hz. Here $\alpha = 1.2$ and $\epsilon = 3$ dB were chosen. One important thing to note is that if passivity is desired (while rendering pure stiffness, i.e. $D = 0$), the mechanical stiffness should be chosen higher than the rendered stiffness [3], [9].

Note that the tradeoff between transparency and torque tracking bandwidth extends beyond impedance rendering: most modern full-body and inverse-dynamics based control methods assume actuators to be pure torque sources, unaffected by joint motion, while this assumption does not hold.

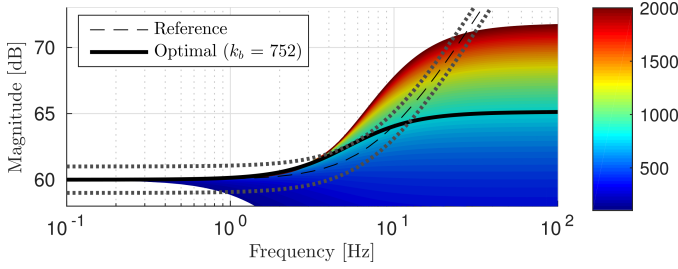


Fig. 8: Impedance rendering optimisation for a reference impedance of $K = 1000$ and $D = 20$. The dotted grey lines show the $\epsilon = \pm 3$ dB acceptable accuracy margin.

V. SIMULATION RESULTS

To validate the analyses presented in Sec. III–IV, this section provides simulation results on the full model shown in Fig. 1, supplemented with nonlinear components, outlined in the next section. The actuators considered in the simulations and experiments are WALK-MAN high-power actuators [13], [15], [17], capable of torques of 300+ Nm. Besides the stiffness values used before, we add two actuators with stiffness values of 2700 Nm/rad (B12) and 11400 Nm/rad (B14), which will also be evaluated in experiment.

We first consider the actuators to be in locked-output configuration to investigate their ability to track desired torques without the influence of a moving load. Then, in Sec. V-C and V-D, we use the same controllers and assess the transparent interaction and impedance rendering performance, respectively.

A. Nonlinear Dynamics

Several nonlinear components were implemented to enhance the realism of the simulations and to show phenomena that do not exist in linear analysis, presented in Sec. III. These were implemented using our Compliant Joint Toolbox [18] for MATLAB/Simulink.

1) *Asymmetric (Coulomb) Friction*: For mechanisms with sliding surfaces Coulomb friction is an important nonlinear damping component which becomes dominant at low velocities. The bearing damping elements d_m, d_g, d_l in Fig. 1 are complemented with asymmetric viscous and Coulomb friction. The friction parameters are identified from experiments on actuator B12 with amplitude modulated pseudo-random binary signals covering the entire position, velocity and acceleration range of the actuator [15].

2) *Controllers*: The controllers are implemented similar to their hardware counterparts (which run on DSPs);

- Discrete-time with 2 kHz sample rate;
- One timestep of output delay;
- Derivative filtering (200 Hz);
- Gaussian current and sensor noise with variances 14 mA and 1.8×10^{-10} rad, respectively;
- Encoder quantisation (19 bits);
- Current limitation (40 A).

3) *Torque Ripple*: There are two primary torque ripple sources present in the system that we take into account:

- Position-dependent detent torque emerging from the electrical motor, acting as a sinusoidal torque on the motor (15 mNm with frequency of $6n$ periods/rev);

- Harmonic Drive torque ripple, which is both position- and torque-dependent, and scales linearly with the torque transmitted through the gearbox ($10^{-4}n \text{ Nm/Nm}$ with frequency of n periods/rev)

where the gearbox ratio $n = 100$.

B. Torque Tracking in Locked-Output Configuration

Fig. 9 shows the closed-loop torque tracking response. As expected, the higher stiffness actuators are able to achieve much higher bandwidth. The controller gains were derived with $\alpha = 2.70, 1.90, 1.50, 1.12, 1.10$ for the 100, 1000, 2700, 10000, 11400 Nm/rad actuators, respectively. Despite using α being a very useful way of choosing gains, selecting a single value of α for very different stiffness values proved impractical due to the involvement of nonlinear dynamics, finite gearbox stiffness, and input saturation. The gains were chosen s.t. input saturation would not occur before -3 dB and no more than +1 dB ripple was observed.

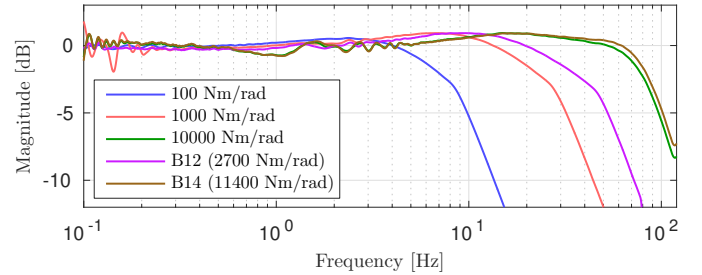


Fig. 9: Simulation: Fixed-output torque tracking.

While the higher stiffness values 10000 and 11400 Nm/rad did not saturate the 40 A input, they see a similar roll-off starting around 60 Hz due to the modelled finite gearbox stiffness. It can be observed that a set of ripples appear at low frequencies. While partly numerical artifacts, they are indeed the result of the nonlinearities outlined in Sec. V-A, and become dominant at lower frequencies and amplitudes.

C. Transparency

We simulate the actuator's transparency by frequency sweeping the output position. As frequency increases, the amplitude is reduced to keep the velocity amplitude constant, to avoid quickly reaching excessive velocities.

Fig. 10 displays the residual torques. As expected based on Sec. III-D, the lower stiffness actuators demonstrate much lower residual torques over the entire frequency range, despite the much lower torque tracking bandwidth. The higher gains for the lower stiffness actuators results in lower residual torques at low frequencies. Furthermore, the residual torques can be seen to converge to k_b (in dB) at high frequencies.

D. Impedance Rendering

We simulate the impedance rendering capability of all five stiffness levels, for a reference impedance of $K = 1000 \text{ Nm/rad}$ and $D = 20 \text{ Nms/rad}$. The results shown in Fig. 11 confirm the validity of the theoretical predictions in Sec. IV; see e.g. Fig. 8. Additional roll-off can be seen at high frequencies, which was confirmed to be attributed to the finite gearbox stiffness.

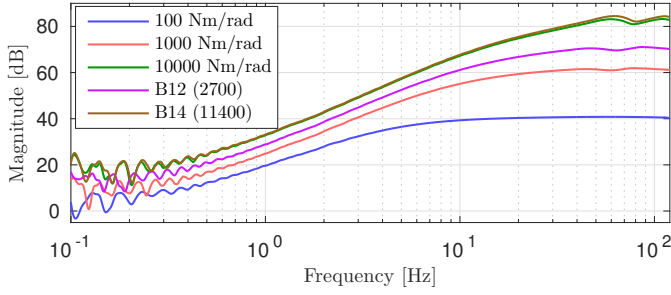


Fig. 10: Transparency ($\tau^* \equiv 0$): Simulated residual torques due to output motion.

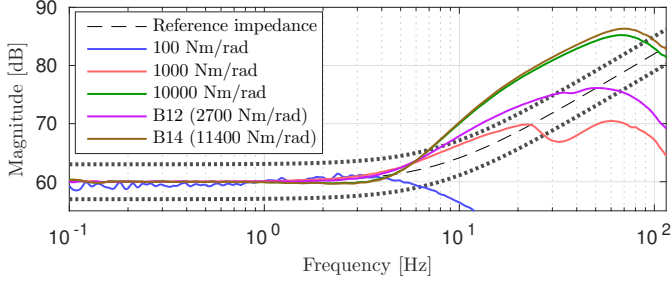


Fig. 11: Simulation: Impedance rendering for a reference impedance of $K=1000$ and $D=20$. The dotted grey lines show the $\epsilon = \pm 3$ dB acceptable accuracy margin.

VI. EXPERIMENTAL RESULTS

A. Experimental Setup

To investigate the influence of actuator stiffness in practice, the previously mentioned actuators B12 and B14 were used. The controllers received torque references through a 1 kHz real-time EtherCAT connection. As in simulation, the torque controller itself was running on DSPs at 2 kHz, and other controller properties were equally unchanged. We include simulated results in the transparency and impedance rendering experiments for comparison.

B. Torque Tracking in Locked-Output Configuration

The actuators were first tested in locked-output configuration, by fixing the output flange to ground. A torque reference frequency sweep was performed, the results of which are shown in Fig. 12. We found the bandwidth of both actuators to increase slower than expected with increased α . However, we used $\alpha = 2.3$ and 3.3 for both actuators in further experiments: the higher gain results in improved low-frequency accuracy (Sec. III-D). Equal values of α for both actuators ensures a fair comparison. The bandwidth for both actuators is lower than expected; while for B12 the natural frequency appears to lie in roughly the expected location based on α , both actuators experience more roll-off than suggested by the simulations. We suspect this is due to nonlinear stiffness and lost motion of the gearboxes.

C. Transparency

To evaluate the actuator transparency, pendulum drop experiments were performed with a 35 cm, 4.5 kg pendulum with an inertia of 0.35 kg m^2 , released from a horizontal position.

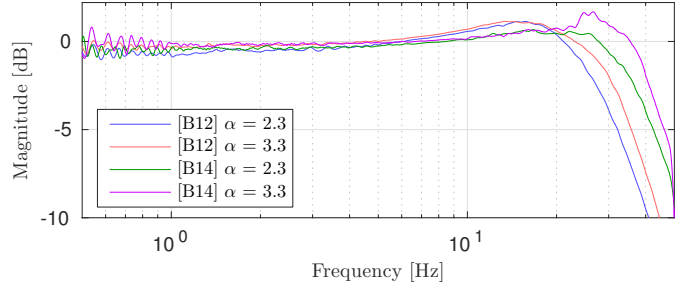


Fig. 12: Experiment: Fixed-output torque tracking.

The results are shown in Fig. 13. As expected, transparency increases with increased gains. However, it can be observed the more compliant B12 offers more transparency than B14 at both gain levels, resulting in a more natural, less damped pendulum swing. This is most likely due to larger friction in B14. Conversely, given that the friction parameters were identified on B12, its simulated and experimental responses are in good agreement. The spike in torque for B12 at 0.45 s is due to reaching the speed limit of $\approx 5 \text{ rad/s}$ imposed by the back-EMF and the operating voltage of 24 V.

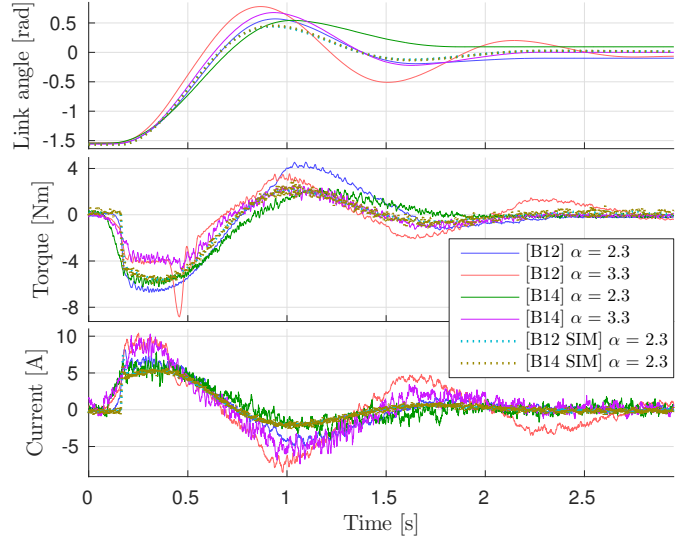


Fig. 13: Transparency ($\tau^* \equiv 0$) experiment: Dropping a 4.5 kg, 0.35 kg m^2 pendulum from a horizontal position.

D. Impedance Rendering

To investigate the impedance replication capability, we set the reference impedance to a pure spring with $K = 100 \text{ Nm/rad}$ and $D = 0 \text{ Nms/rad}$. The pendulum was moved from a hanging position by 20° and released. As shown in Fig. 14, similar to the pendulum drop findings, the lower stiffness actuator shows larger capability of handling the load motion, resulting in a lesser damped response. While this can in part be explained by differences in friction, the difference was observed to increase at larger impedance values (not shown for brevity); at $K = 1000 \text{ Nm/rad}$ the resonance frequency with the pendulum is 8.5 Hz, well below the torque control bandwidth for both actuators. Therefore, the transparency function is a limiting factor in the actuator performance.

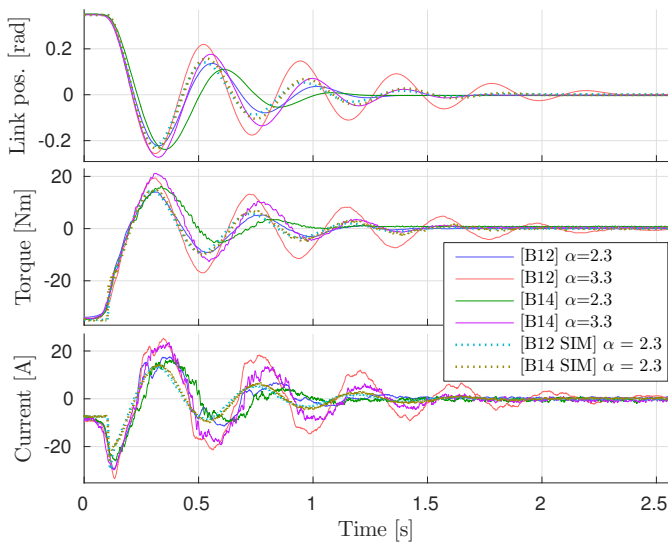


Fig. 14: Experimental impedance rendering for $K = 100 \text{ Nm/rad}$ and $D = 0 \text{ Nms/rad}$.

VII. CONCLUSIONS & FUTURE WORK

This paper has presented an extensive analysis of the aspects that influence torque control and impedance rendering performance of SEAs. Particularly, stiffness selection and control parameters were discussed in-depth.

Through the dichotomy of its dynamics, it was shown that generated torque is the sum of input responses from torque reference and load motion. Several analyses were consolidated to study the effects of mechanical compliance and control parameters on the performance of SEAs in terms of tracking zero and non-zero torques in the presence of load motion, as well as active impedance rendering. Two opposing criteria exist for the choice of stiffness: lowering stiffness increases transparency, and increasing stiffness increases achievable torque tracking bandwidth. This fact was used in the context of impedance rendering to propose an optimal stiffness selection criterion, which maximises the frequency up to which an impedance can be rendered accurately. This criterion allows to select the actuator stiffness with respect to force and motion control performance requirements dictated by the targeted robot and field of application. Real-world design is a compromise between many criteria, such as impact resilience, volume, and force/motion control performance.

Results from simulations and experiments were presented to validate the analyses. Particularly, it was shown the stiffness has a significant effect on the renderable impedance values and accurate rendering bandwidth. Experiments showed higher transparency of the more compliant actuator even at very low frequencies, which was unexpected, but may be explained by differences in friction. The analyses and simulations correctly predicted the transparency and impedance rendering dependence on gains. The results from both simulation and experiment confirm our theoretical analyses. Comparing impedance rendering capability in practice proved difficult; using a secondary motor driving the actuator externally should provide further insight and is left for future study.

While the actuators considered in this paper are those with linear stiffness, we are particularly interested in nonlinear

(stiffening) profiles, as they may combine the low stiffness optimal for transparency with the high stiffness optimal for torque tracking. This will be considered in future works.

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